

Physics-Informed Machine Learning for Robust Building Energy Prediction under Incomplete Sensor Data

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Abstract

The application of smart building prediction models is critically reliant upon streams of stream data collected from devices that often fail, are unable to communicate, have calibration issues or use poor sensor coverage. This model proposes to use a physics-informed machine learning model to capture predictive building energy models with missing sensors. The study will utilize the public energy, HVAC, indoor outdoor environmental and occupancy data for the last complete 3 years of operation for an office building. Gaps in the points and outages for contiguous sensors will be added at 10% to 40% rates randomly. A simplified model of electrical resistance and capacitance heat balance, which is also mask-aware, will be used to regularise this neural model and will be compared with persistence, XGBoost, LSTM, and an architecture-matched non-physics model. The accuracy of performance will be checked based on MAE, RMSE, R squared, error degradation, physical residuals, repeated seed confidence interval. Expected contribution: implement ability to run the robustness benchmark in a reproducible way, and evidence of when the physical knowledge improves the predictions in the environments where sensors on buildings are providing poor measurements.

Keywords: physics-informed machine learning, building energy prediction, missing sensor data, smart buildings, robustness

1 Introduction

As technology advances, times change, buildings are now beginning to be equipped with digital meters and sensors observing the electricity needs in the premises, the temperature and air temperature, the humidity, occupancy and whether the heating, ventilation and air-conditioning (HVAC) system is working. These data can be used for forecasting purposes, and for fault detection and control decisions, but models for their prediction may not be robust in the field since they

may assume optimal laboratory conditions. Only a few building-management systems offer ideally accurate data: sensors can break down, networks can lose data, measurements drift and some variables can simply not be measured due to high installation cost or in a way that is intrusive. A model that only sees complete data records can then make unreliable and/or physically implausible predictions when an operator needs reliable guidance.

Purely data-driven machine learning is aimed at the learning of nonlinear relationships between weather, occupancy, thermal conditions and energy demand; it doesn't necessarily model the concepts of heat storage, heat transfer, and energy balance. The model might correlate some observations which derive from non-existing observations in the following time span. One way to overcome this limitation is to make use of prior physical knowledge in the architecture, features, or training objective, as Physics-informed machine learning (PIML) does. The approach does not involve replacing data with a full engineering simulation, rather it initiates modeling with some relationships that are only approximate and then restricts the range of acceptable models' predictions. It is especially applicable to buildings where a simplified RC model can be used to describe the time response between the indoor temperature, outdoor temperature, thermal gains from the building and HVAC power.

The literature reveals that PIML is an efficient way to leverage more data, predict thermal issues at longer time horizons, and ensures physical consistency among data, and missing-data studies in recent literature demonstrate the impact of imputation method may be significant for the performance of building-energy models. Typically two streams of research have been considered independently however. The typical building PIML studies assume that the appropriate operational sensors can be found while the typical imputation studies tend to typically optimize for reconstruction or benchmark accuracy without considering whether there exists a physics-constrained predictor that manifests improvement of sensitivity to failure compared to a conventional predictor. The question of whether PIML can model buildings in general becomes the question of PIML's physical prior giving measurable protection with multiple operational inputs incomplete.

Research question. What are the advantages of physics-informed machine learning over conventional machine learning models in predicting building energy use when data is not fully available from sensors?

The three sub-questions are: (RQ1) How does prediction accuracy change when various percentages of input data are missing; (RQ2) Does the accuracy vary between missing readings in a random order and missing readings in contiguous order of the readings; and (RQ3) Will greater accuracy benefit come with lower physical balance violations and explainable effective thermal parameters.

Proposed solution and objectives. The mask-aware neural predictor will jointly estimate HVAC energy and the next status of the indoor temperature. The loss function will be made up of prediction error and a simplified thermal energy-balance residual. The implementation will advocate four goals:

- Create a repeatable benchmark of incomplete sensors fixing the splits in a reproducible manner and the equivalent masks missingness for each model.
- Follow standards for implementing conventional baselines and model using learning effective thermal coefficient in a physics-informed model.
- Compare evaluation of accuracy, robustness, physical consistency, uncertainty and computational cost of the algorithms under controlled missingness conditions.
- Deliver usable recommendations for when physics-informed modelling should be useful, neutral or unhelpful for the missing data situation for a specific real-world problem.

The expected contribution is neither theoretical nor a model of the entire building, but an empirical and methodological one. The work will begin by establishing a link between physics-informed prediction and more realistic sensor-loss experiments for the first time. Second, an architecture matched ablation will control for the effect of network capacity to help isolate the effect of the physics term. Third, the data-processing, masking and evaluation pipeline will be published and used for comparisons in future studies. The outcome of the engagement is evidence to support facility decision makers in evaluating if they have sensor coverage to cover all their facility needs and if they do not, if the physical restraints are worth the implementation costs.

Following this, the remainder of the proposal is structured so that. Section 2 critically reflects on the issues of building energy prediction and the treatment of missing data in building modelling, followed by identification of the main niche of building-reflecting basic study and research. Section 3 outlines experimental design, dataset, proposed model, baselines and evaluation strategy, resources, ethical considerations and risks as well as project plan.

2 Literature Review

2.1 Building-Energy Prediction and Incomplete Operational Data

There are two measure that are useful for considering a good predictor of use of energy: how well it predicts the energy on testing data and how the size of the error varies if one departed from the same data at deployment as the one used for training. Luo et al. (2022) illustrate the richness and complexity of real building data, showing that a three-year office-building dataset of over 300 energy, HVAC, indoor environment, outdoor environment and occupancy classes consists of sensors and meters. While their curation pipeline’s purpose is to close the data gaps and make research data, the missing rate documentation that they used when the system was originally designed indicates that the process of seeking out missing data is an intrinsic property of the system and is not an artificially created ‘problem’. Kim and colleagues (2023) also make a comparison

of imputation techniques for the recovery of missing measurements, but they focus on residential energy, behaviour and environmental variables. The present research question is not addressed by either of the studies, since the same sensors are removed from both types of predictors (a downstream physics-informed and a standard predictor) to determine which is more robust.

The missingness mechanism is also consequential, as is assumed. While Kim et al. (2023) discuss other types of missing observations, Lee et al. (2024) claim that building data are often better thought of as missing at random (MAR) than completely at random (MCAR). Lee et al. contrast how median and K-nearest-neighbour, CART and XGBoost handle the imputation and demonstrate that the data distribution reconstructed and the stability of the resulting benchmarking model during the imputation can change depending on the imputation method used. Their result contradicts research which only reports a single randomly masked test set. The proposed research therefore attempts to distinguish outages from just a missing point from contiguous outages and includes valid masks with random initial seeds and the target energy series is observed, so that the experiment gauges robustness to outages in the input, not mixed outages with unknown ground truth.

This "missing-data" research provides robust "conventional" baselines, but does not consider the physical relationships primarily as a constraint in training. Linear Interpolation and CART-based imputation are retained because these are transparent, addition of CART allows for graphing prior to building, and both can sufficiently and efficiently interrelate with missing data, as well as have prior building data comparisons that support their use. XGBoost is added to properly and effectively interrelate with missing data and has efficient handling of nonlinear tabular relationships. They are limiting and motivate the next subsection: An imputed value could be statistically plausible, but fall short with regard to temporal heat balance, which relates temperature change and HVAC energy.

2.2 Physics-Informed Models for Building Thermal Dynamics

Physics-informed machine learning can be implemented in various ways, including by means of soft penalties, constrained architectures and hybrid physics-informed data modules. Physical laws can provide information beyond observed labels, but optimisation is complicated if there are losses in the data and physics that have incompatible scales, explains Karniadakis et al. (2021). In the building sector, the thermal dynamics are encoded in order to predict the room temperature, energy consumption and an unobserved thermal-mass state, as conducted by Gokhale, Claessens and Develder (2022). Their models need less training data than traditional neural networks, and can accurately predict the future for longer periods of time. This reinforces the idea that physics can be of aid when measurements are lacking, but does not imply the same as not having sufficient training data, since it is the latter that differs when encountered at runtime with inference: it changes the set of variables available for action.

In a different approach, Di Natale et al. (2022) come up with the ideas of a Physically Consistent Neural Network, which is physical in both sense of the term: first by the fact that its structure reflects a sensible directional response to the heating inputs, and secondly because the structures allow to simultaneously account for the neighbouring temperatures and the outside temperatures. Their model can achieve the accuracy of neural networks, eliminate overfitting and enhance the physical consistency. Similarly, a thermal model and a neural network are also integrated into PhysCon, which is a framework for thermal modelling and demand response control, as proposed by Chen et al. (2023). The structural constraints can offer more rigorous guarantees but unfortunately less flexibility if the dataset is missing the exact level of power or building parameters at the zone level. As a result, the present work adopts a weak residual, with learnable physics weight, which is an inferior form of guarantee, but much more practical for a public operating data set, and in which the effect of the physics weight can be validated by ablation.

In all these studies, there have been some common advantages for using improved data efficiency and physical plausibility and limitation for transferring between buildings, climates and sensor configurations. Ma et al. (2025) identify the balancing of accuracy and physical consistency, the selection of physics priors and real-world validation as open issues in building PIML. The aforementioned problems are directly linked to the research question, with an over-strong but underspecified thermal equation increasing the potential of errors to arise if missing data were to be considered. The proposed model will thus learn the thermally effective coefficients when they are constrained to be positive, and will only fine-tune the weight of physics loss on the validation set without exploring the notion of a better and better regularisation, as it will instead issue both predictive error and physical loss as residual.

2.3 Physics-Informed Imputation and Energy-Efficiency Prediction

The most recent relation between building physics and missing data is provided by Liguori et al. (2024). Their physics-driven denoising autoencoder couples together the temperature, heating and cooling reconstruction with a thermal-balance equation. There was very little increase in reconstruction error for the physics term, but there was improved robustness over missing data rate, and physics term coefficients were interpretable. This is crucial as it demonstrates that the primary advantage of physics could be graceful degradation or interpretability, not the least error for full data and for partial data. The study, however, looks at imputation as the goal. The proposed work is instead to investigate the use of these energy inputs to predict energy downstream and then compare the full model with the “zero physics weight” model created for the physical platform.

Michalakopoulos et al. (2024) suggest using a physics-informed neural network to develop a building energy-efficiency prediction model for 256 residential buildings. The network’s outputs are an estimate of building-envelope and ther-

mal properties, which then are fed into heat-loss equations for use the rest of the way through the energy calculation and training objective. Unlike Liguori et al. (2024), the study aims at energy performance instead of reconstruction in time series and the input are the characteristics of the audited stationary sensor, while in Liguori et al. (2024) the sensors are removed systematically to reconstruct the time series. Both papers set up a baseline physics-informed energy-prediction approach, neither integrating operational time-series energy prediction with controlled sensor outages. Both set up a baseline physics-informed approach to energy prediction, none of which integrate operational time-series energy prediction with controlled sensor failures.

2.4 Principal Baseline Study and Research Niche

Principal baseline study. Of the papers reviewed, the principal baseline study is of Michalakopoulos et al. (2024) as the authors' study most directly shows how physics-informed building energy prediction can be used in publications. The main features of the proposed study include the continuation of the essence of the previous problem of estimating residential energy-efficiency but transition of problem from static to operational energy forecasting in an office building and introduction of systematic incomplete-sensor experiments to the data-driven network. The complementary benchmark that is missing data is utilized to design the masking, ablation and robustness analysis is taken from Liguori et al. (2024).

The research niche is thus a controlled experiment to understand if approximate building physics can decrease prediction-error degradation when the inference time sensors are not present. The anticipated novelty is rooted in the four following points: Identical masking of missingness in all models; Both point outage and block outage scenarios; No-physics outage (to account for robustness slope and physical residual) matched to model architecture; Joint reporting of predictive accuracy, robustness slope, and physical residual. Fits within a 'narrow but still significant' niche that is small enough for a capstone implementation but still provides literature documented evidence that is required for successful real-world PIML adoption.

Table 1. Critical evidence map for the proposed research

Study group	Primary focus	Strength for this proposal	Limitation creating the niche
Luo et al. (2022)	Rich operational dataset	Real energy, HVAC, environment and occupancy sensing	Cleaned data require synthetic, controlled re-masking
Kim et al. (2023); Lee et al. (2024)	Missing-data handling	Imputation choice and missingness mechanism affect results	No physics-informed downstream prediction comparison
Gokhale et al. (2022); Di Natale et al. (2022)	Physics-guided thermal models	Data efficiency, long-horizon accuracy and consistency	Sensor-outage robustness not directly evaluated
Chen et al. (2023); Ma et al. (2025)	Building PIML design	Hybrid models are promising; physics weighting and validation remain open	Potential model misspecification and limited transfer
Liguori et al. (2024)	Physics-informed imputation	Robustness and interpretable coefficients under missingness	Imputation, not energy prediction, is the endpoint
Michalakopoulos et al. (2024)	Physics-informed energy prediction	Direct energy-efficiency baseline	Static audited features; no controlled sensor loss

3 Research Method & Specification

3.1 Research Design and Justification

The research to be carried out will be a quantitative computational experiment (design-science implementation). It is a suitable design because the research question is a measurable one based on comparisons of an artefact to known alternatives in controlled settings. A pure case-study would describe a single deployment and a pure theory study would determine if the proposed constraints would be helpful if used with operational data containing a lot of noise. The experiment thus uses varying missingness rates, generation of outage patterns, fixed train-test and model-selection budgets, and compares the accuracy and physical consistency of the results.

Type of model and missingness rate and missingness pattern will be independent measured variables. dependent variables will be MAE, RMSE, R-squared, Normalised RMSE, Error degradation because of the lack of complete inputs, mean squared physical residual, training time and inference time. The two main hypotheses are: (1) the full physics-informed model will have a smaller RMSE increase compared with the strongest conventional hypothesis at 20%-40% missingness, and (2) the largest conventional hypothesis on earth at 20%-40% missingness will have a larger RMSE increase compared with the full physics-informed hypothesis. A secondary hypothesis is that it will minimize the amount of physical residuals and does not significantly impact accuracy of complete data. These hypotheses are testable and these hypothesis do not turn out to be negative or null in case the physics prior is incorrect or not superior.

3.2 Dataset, Scope and Pre-processing

The primary dataset will come from the public dataset of Building 59 described by Luo et al., (2022). It measures electricity, whole-building HVAC operating variables, environmental variables, and occupant counts, for a three-year period of a Berkeley office building. Analysis of the two office floors will be done with an extracted subset of documented data while hourly sampling of both horizontally aligned floors and hourly predictions of the next hour HVAC electricity demand will be performed. Lagged HVAC energy, aggregated indoor temperature, outdoor temperature and humidity, occupancy, selected HVAC operating signals and calendar variables are candidate inputs. To avoid “dressing up” the signal, the actual signal will be recorded in a data dictionary prior to modelling.

The split by using one for training, one for validation, and one for testing will prevent information getting lost across the years: 60% for training, 20% for the validation period, and 20% for the testing period after the pandemic. Model tuning will not be done for the regime-shift year of 2020 since there was a significant change in occupancy and operating patterns, so the year will be left as an external regime-shift. Statistics to be used in scaling and imputation are only training set statistics. The dataset documentation will flag values that are impossible by themselves and any natural gaps that existed prior to the experiment will be noted without being discarded (silently removed).

Input sensors will only be controlled missed after chronological split. If any readings are removed the MCAR scenarios will, by themselves, remove the reading, while block-outage scenario will remove a range of contiguous readings across 1 or more sensors. The 5 fixed seeds will produce 10%, 20%, 30% and 40% rates. The same masks will be given to all models. Each sensor will be accompanied by a binary observing mask, as well as a time since last observing sensor value, to differentiate an actual zero value from the lack of one. In addition to writing out the following explicit baselines for “pre-processing,” linear interpolation and CART imputation will be used, with no imputed test value taken from future observations outside the forecastable horizon.

3.3 Proposed Physics-Informed Model

The proposed artefact is a mask-aware recurrent or temporal multilayer neural network, two outputs of which will be the energy consumed by the HVAC system and the representative indoor temperature for the next hour, respectively. The data branch will be trained with some of the same features fed to the neural baseline, but will be tasked with finding non-linear temporal relationships. In short, a branch of physics will incorporate a somewhat simplified first order resistance-capacitance balance for thermal storage, heat exchange with outdoors, internal/occupancy gains and the power needed for HVAC equipment. Effective thermal capacitance, conductance and gain will be learned in a way that respects positivity constraints, instead of relying on absent construction drawings.

$$r.t = C(T_{\text{hat}}[t+1] - T[t]) / \Delta.t - \{UA(T_{\text{out}}[t] - T[t]) + \eta P_{\text{hat}}[t] + \beta O[t] + q_{\text{solar}}[t]\}$$

Here α is set to 1.5, λ is 0.5, and γ is 5.0, determined in the earlier experiments described. Parameters α , λ , γ are set to 1.5, 0.5 and 5.0, respectively, as determined in the previous experiments mentioned.

Robust terms for energy loss and temperature will be applied to the energy and temperature losses, and a robust physical term for residual will be applied to those predictions that are unable to be matched with the simplified heat balance. The degree of physics was expressed by the parameter λ . It will only be tuned on validation data, in a prespecified logarithmic range. The choice of this soft constraint is undertaken due to the possibility that not all of the zone model’s parameters are available from the public data set. It has the benefit of being feasible and it doesn’t require exact physics to do that, but it is also limited by how well the predictor performs, as well as by excessive λ or poor residual values. To avoid this pseudo-identification, a sensitivity analysis has been performed on the study and the learned parameters reported in the form of ranges, instead of an absolute value.

3.4 Baselines, Ablations and Fair Comparison

- Seasonal persist: Hour/day energy from previous season, offers a clear base level.
- Strong nonlinear tabular baseline: XGBoost using selected imputation / mask features.
- LSTM with same imputation and mask features, and temporal windows. • gated recurrent network with the same temporal windows, imputation and mask features.
- Isolation of the influence of physical regularisation from the network capacity using architecture-matched neural model with $\lambda = 0$.
- This includes a full physics-informed model, as well as models in which all the mask terms or all the physics terms were removed to leave only the auxiliary tasks.

The same chronological data and the same target are used for all trainable models; the same prediction horizon and missingness masks; and the same early-stopping rule. A fixed number of validation trials per model will be used for hyperparameter search in order to not give the physics-informed approach an unfair optimisation budget. The final test set will be tested once after model selection. A primary disadvantage in comparative studies is introducing unequal tuning or different data preparation which can be interpreted as a methodological benefit. Different tuning or data preparation is a common issue in comparative studies where it seems that a methodical advantage was implemented; this design resolves that issue.

3.5 Evaluation and Evidence

The absolute prediction error will be measured by the MAE and RMSE terms, explained variance by R-squared, and normalised RMSE will allow comparisons to be made over periods of varying loads. The measure of robustness will be represented by the percentage increase in RMSE (measured as percentage from complete data) and the slope (gradient) of the error of missingness rate curve. The distance from each cure’s passively restored thermal equation to the set of correct equations would be measured with the help of mean squared thermal residual and the percentage of predictions making the implausible negative thermal energy. Training/Inference time will record the price of the addition of physics.

Results will be presented separately for MCAR and block outages and in terms of each level of missing. There will be 5 matched model outputs; 95% bootstrap confidence intervals will be calculated for differences among the metrics; a paired non-parametric test will be performed where distributional assumptions are not believable. In addition, error will also be stratified by occupied/unoccupied hours and the mild/extreme outdoor-temperature periods. The success criterion is estimated to be at least a 5% reduction in the RMSE from the best conventional baseline at one or more of the missingness rates from 20% to 40% prior to the submission and with no more than 2% higher RMSE in the complete data setting. The threshold is designed to be used for interpreting this; it will however not be applied to suppress contrary results.

Paired error curves across missingness rates, an architecture matched physics ablation, robustness on contiguous outages, coefficient and physics-residual analysis, and reproducible code with fixed seeds and data splits will be provided as the key evidence to answer the research question. That difference will be reported if it appears only when the model is run using the imputations, but not when run using the original masks. The contribution will be eligible to be presented as that for a season or regime only if this contributes to its gain rather than providing a general contribution.

3.6 Research Resources, Reproducibility and Risk Control

Table 2. Planned research resources and controls

Resource	Planned use	Quality/reproducibility control
Data	LBNL Building 59 public dataset; documented subset and data dictionary	Licence and citation checked; raw data retained read-only
Software	Python, pandas, NumPy, scikit-learn, XGBoost, PyTorch, Optuna	Pinned environment and deterministic seeds
Compute	4-8 CPU cores, 16-32 GB RAM, optional single T4/A10-class GPU, 50 GB storage	Pilot on CPU; cap search trials and window size
Tracking	Git, experiment configuration files, metric logs and generated masks	One command reproduces each table/figure
Validation	Unit tests for leakage, masking, inverse scaling and metric calculation	Manual spot checks against raw time series

The main drawbacks are complexity of the datasets, misrepresentation of the thermal prior and overcomplicated missingness. These are all to be channeled through starting on one documented subset of floors/end-use, comparing multiple physics weights such as zero, testing with two different outage scenarios (MCAR and contiguous), etc. and mastering a small pilot project before running an exhaustive hyperparameter search. Another risk of validity concerns the lack of universal transfer in one building in a mild climate. The report will thus separate out internal strength from cross building generalisability and identify multi building replication as future work.

3.7 Ethical Considerations of the Research

No human subjects will be recruited, and no additional personal information will be collected, and no human subjects will serve as subjects in the study. However, the number of people that occupy a site can provide clues to the presence of occupants. The analysis will be based on counts only, not be made to seek re-identification and not attempt to string the data together with data from other identification sources. Documentation about the dataset licence conditions, the requirement for attribution and the requirements for the repository will be documented in the project documentation. Any raw data would be kept securely and a non-identifying result list will be incorporated into the report.

The responsible modelling also should not include deceptive or misleading operational claims. A physics-informed prediction cannot be relied upon simply to be safe or accurate, especially if the physics prior involves some simplifications. The model will thus be presented to be used as a decision support tool and not an independent HVAC controller - limitations of the model will be reported, and if measurements are lost severely, uncertainty/error will be apparent. Pilot runs technique, early stopping and bounded hyperparameter searches will be adopted in computational experiments for avoiding unnecessary energy use. The ethics declaration will make it explicit if public secondary

data is used, occupancy variables are used, cloud computing or generative-AI (gen-AI) assistance has been utilized, as mandated by NCI policy.

3.8 Project Plan

A twelve weeks capstone schedule is used for planning a work. The modeling has explicit dependencies; full modelling only starts after the data dictionary and leakage tests, the physics modelling is based on validated baselines, and the final writing is done after the experimental protocol is locked. Once the capstone calendar is finalized, the dates for the capstone will be entered in editing the Gantt chart below.

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Protocol, ethics and literature												
Data acquisition and dictionary												
Cleaning, leakage tests and EDA												
Missingness and imputation												
XGBoost/LSTM baselines												
PIML model implementation												
Tuning and main experiments												
Ablation and statistical tests												
Interpretation and figures												
Report, code and viva												

Figure 1. Twelve-week editable project Gantt chart

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- Venue-quality note: CORE ranks conferences rather than journals. Journal quartiles below are reported using JCR; the conference paper is reported using CORE availability and a dated Google Scholar citation count.
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